Q4 Reflection: PRD Genie

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1. What the traces showed

Langfuse traces of one full chain (Analyze, PRD, Stories) showed where time and money go. The chain took 36 seconds and cost about $0.0022. Stories took 53% of the time and 55% of the cost. Output tokens were 25% of all tokens but 58% of the cost, so the useful lever is making agents write less, not read less. The Extractor's fixed rules were about 97% of its input (1,375 tokens, of which about 35 were the user's text), so a vague input costs little more than a detailed one. It takes longer only because the Gap agent writes more questions (12.1 s for T2 against 9.1 s for T1). About 23% of the wall-clock time sat outside the model calls.

Reading runs side by side also showed something a single run hides: the same input did not always give the same result at temperature 0. One T6 run flipped to INSUFFICIENT, and one T4 PRD lost its persona.

2. What the evaluation loop taught me

The loop was evaluate, find a failure, fix the prompt, and re-evaluate. The Stories agent first drafted defaults and message texts nobody had stated and paraphrased the acceptance criteria. After a stricter prompt, T4, T8 and T12 passed with criteria copied verbatim and no invented specifics. T9 first returned an empty result with no explanation. A rule that INSUFFICIENT must name what is missing made the Gap agent ask for it.

Three lessons stand out:

- Keyword checks and human judgement measure different things. T7 meets its criteria, yet a PM reading it asks which API is meant, and the thin stories also pass. Across nine annotated traces, faithfulness averaged 4.9 out of 5 and completeness 4.7, but usefulness only 3.3: the agents rarely invent, but they sometimes fail to flag or capture something.
- A rubric must say what is scored. My first annotation scores marked vague inputs down for being vague. The right question is whether the output handled the input correctly.
- One passing run is not proof. Fixing one thing exposed another: removing a visible flag let the model invent a benefit until a rule closed that gap. A fixed baseline set must be re-run after every prompt change.

3. Risks of AI-generated PRDs

- Fabricated requirements that engineering then builds. Mitigations: an exact-quote rule, a sufficiency gate, and draft tags.
- A polished document earns more trust than it deserves, so a PM may approve without reading. Mitigations: approval between stages, a review status on every story, and Not specified in source markers.

- Missed conflicts and gaps, which fail silently (T3).
- Variation between runs, which makes one good result misleading.
- Confidential meeting content sent to a third-party model.
- Accountability: the PM, not the agent, owns whatever is signed off.

4. Improvement plan

1. A questions-first step: ask the PM decision questions, each with a proposed default, before stories are written, so business rules and validations come from answers. This needs a custom UI, because Langflow's Human Input component offers buttons only.
2. Teach the Gap agent to ask which one is meant when the input names something without identifying it (which competitor, which API, which dashboard). This appeared in T2, T5 and T7.
3. Add a conflict check for implicit trade-offs (T3).
4. Reduce variation: validate each model's output against the rules in code and retry, and run each baseline test three times.
5. Automate evaluation: an LLM judge checked against the PM annotations, and the baseline set as a regression run before every prompt change.
6. Run a pilot with 3 to 5 PMs to measure time saved, which the traces cannot show.